

Model policy vs sweep-best policy — mixed workloads

Current model after early-diversity change. Gap = 100 × (model time / best time − 1).

Workload	Worker budget	Model allocated policy (modal if varying)	Model time (ms)	Sweep-best policy (remeasured)	Best time (ms)	Extra time (ms)	Gap from best (%)
Mixed: compute	4	Dynamic (3, 3) P=1; order=LRabc; BF=0 * 80% of native samples	0.9393	Dynamic (3, 3) P=1; order=RabcL; BF=0	0.9299	+0.0094	+1.0%
Mixed: compute	6	Dynamic (5, 5) P=1; order=LRabc; BF=0 * 80% of native samples	0.6665	Dynamic (5, 5) P=1; order=LabcR; BF=1	0.6810	-0.0145	-2.1%
Mixed: long ordinary	4	Dynamic (1, 1) P=3; order=abcLR; BF=0 * 60% of native samples	0.6321	Dynamic (1, 1) P=3; order=LabcR; BF=1	0.6408	-0.0086	-1.3%
Mixed: loop skew	4	Dynamic (3, 1) P=1; order=LRabc; BF=0	0.2683	Dynamic (3, 1) P=1; order=RabcL; BF=1	0.2688	-0.0004	-0.2%
Mixed: loop skew	6	Dynamic (5, 1) P=1; order=LRabc; BF=0	0.2188	Dynamic (5, 1) P=1; order=aLRbc; BF=1	0.2140	+0.0048	+2.3%
Mixed: medium ordinary	4	Dynamic (1, 1) P=1; order=LRabc; BF=0 * 40% of native samples	0.2705	Dynamic (3, 3) P=1; order=LabRc; BF=0	0.2518	+0.0188	+7.5%
Mixed: medium ordinary	6	Dynamic (1, 1) P=1; order=LRabc; BF=0 * 60% of native samples	0.2792	Dynamic (5, 5) P=1; order=abRcL; BF=1	0.3289	-0.0497	-15.1%
Mixed: one long ordinary	4	Dynamic (1, 1) P=2; order=cLRab; BF=0 * 60% of native samples	0.4245	Dynamic (1, 1) P=2; order=LcRab; BF=1	0.4121	+0.0125	+3.0%
Mixed: one ordinary	2	Dynamic (1, 1) P=1; order=LRa; BF=0	0.2248	Dynamic (1, 1) P=1; order=aLR; BF=0	0.2239	+0.0009	+0.4%
Mixed: one ordinary	4	Dynamic (3, 3) P=1; order=LRa; BF=0 * 80% of native samples	0.2235	Dynamic (3, 1) P=1; order=aRL; BF=0	0.2173	+0.0062	+2.9%
Mixed: one ordinary	6	Dynamic (1, 1) P=1; order=LRa; BF=0 * 80% of native samples	0.2277	Dynamic (5, 1) P=1; order=aLR; BF=1	0.2133	+0.0144	+6.8%
Mixed: short ordinary	2	Dynamic (1, 1) P=1; order=LRabc; BF=0 * 80% of native samples	0.2244	Dynamic (1, 1) P=1; order=aLbcR; BF=1	0.2221	+0.0023	+1.0%
Mixed: short ordinary	4	Dynamic (1, 1) P=1; order=LRabc; BF=0 * 40% of native samples	0.2264	Dynamic (3, 1) P=1; order=RLabc; BF=0	0.2173	+0.0091	+4.2%
Mixed: short ordinary	6	Dynamic (1, 1) P=1; order=LRabc; BF=0 * 60% of native samples	0.2526	Dynamic (5, 1) P=1; order=aRLbc; BF=1	0.2406	+0.0120	+5.0%
Mixed: two ordinary	2	Dynamic (1, 1) P=1; order=LRab; BF=0 * 60% of native samples	0.2334	Dynamic (1, 1) P=1; order=aLRb; BF=0	0.2328	+0.0006	+0.2%
Mixed: two ordinary	4	Dynamic (3, 3) P=1; order=LRab; BF=0 * 60% of native samples	0.2420	Dynamic (3, 1) P=1; order=aLRb; BF=1	0.2871	-0.0451	-15.7%
Mixed: two ordinary	6	Dynamic (1, 1) P=1; order=LRab; BF=0 * 60% of native samples	0.2310	Dynamic (5, 1) P=1; order=LRab; BF=0	0.2550	-0.0240	-9.4%

Policy: (left, right) loop widths; P = parent-worker limit; order = task priority; BF = backfill (1 on, 0 off).

L/R = loops; a/b/c = ordinary tasks. Priority order is not a serial execution sequence. Width 1 runs inline.

Times: median of process medians. Native time includes policy overhead. * = varying policies; time includes all native choices.

Best is the prior exhaustive-sweep winner, remeasured here. No new full sweep; negative gaps can reflect timing variation or a better native policy.

Model policy vs sweep-best policy — loop workloads (1/3)

Current model after early-diversity change. Gap = 100 × (model time / best time – 1).

Workload	Worker budget	Model allocated policy (modal if varying)	Model time (ms)	Sweep-best policy (remeasured)	Best time (ms)	Extra time (ms)	Gap from best (%)
balanced compute	2	Dynamic (1, 1) P=2; order=LR; BF=1 * 80% of native samples	2.7539	Ordinary-first (1, 1) parents=2	2.6092	+0.1447	+5.5%
balanced compute	4	Dynamic (3, 3) P=1; order=LR; BF=1	1.9895	Dynamic (3, 3) P=1; order=RL; BF=1	1.9505	+0.0390	+2.0%
balanced compute	6	Dynamic (5, 5) P=1; order=LR; BF=1 * 80% of native samples	1.2485	Dynamic (5, 5) P=1; order=RL; BF=1	1.2408	+0.0077	+0.6%
balanced compute	8	Dynamic (7, 7) P=1; order=LR; BF=1	1.4025	Dynamic (5, 6) P=1; order=LR; BF=1	1.2097	+0.1928	+15.9%
chunk edge	2	Dynamic (1, 1) P=1; order=RL; BF=1	0.0131	Dynamic (1, 1) P=1; order=RL; BF=1	0.0122	+0.0009	+7.0%
chunk edge	4	Dynamic (1, 1) P=1; order=RL; BF=1	0.0131	Dynamic (1, 1) P=1; order=RL; BF=1	0.0122	+0.0009	+7.2%
chunk edge	6	Dynamic (1, 1) P=1; order=RL; BF=1	0.0131	Dynamic (1, 1) P=1; order=RL; BF=1	0.0122	+0.0009	+7.1%
chunk edge	8	Dynamic (1, 1) P=1; order=RL; BF=1	0.0132	Dynamic (1, 1) P=1; order=RL; BF=1	0.0124	+0.0008	+6.7%
large arrays	2	Dynamic (1, 1) P=2; order=LR; BF=1	6.1227	Ordinary-first (1, 1) parents=2	6.0348	+0.0878	+1.5%
large arrays	4	Dynamic (3, 3) P=1; order=LR; BF=1	5.4748	Dynamic (3, 3) P=1; order=LR; BF=1	5.3319	+0.1429	+2.7%
large arrays	6	Dynamic (5, 5) P=1; order=LR; BF=1	5.3150	Dynamic (5, 3) P=1; order=LR; BF=1	5.6262	-0.3113	-5.5%
large arrays	8	Dynamic (7, 7) P=1; order=LR; BF=1	7.1927	Dynamic (5, 3) P=1; order=RL; BF=1	5.5440	+1.6487	+29.7%

Policy: (left, right) loop widths; P = parent-worker limit; order = task priority; BF = backfill (1 on, 0 off).

L/R = loops; a/b/c = ordinary tasks. Priority order is not a serial execution sequence. Width 1 runs inline.

Times: median of process medians. Native time includes policy overhead. * = varying policies; time includes all native choices.

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Model policy vs sweep-best policy — loop workloads (2/3)

Current model after early-diversity change. Gap = 100 × (model time / best time − 1).

Workload	Worker budget	Model allocated policy (modal if varying)	Model time (ms)	Sweep-best policy (remeasured)	Best time (ms)	Extra time (ms)	Gap from best (%)
short	2	Dynamic (1, 1) P=1; order=LR; BF=1	0.0456	Dynamic (1, 1) P=1; order=LR; BF=1	0.0442	+0.0014	+3.2%
short	4	Dynamic (1, 1) P=1; order=LR; BF=1	0.0452	Dynamic (1, 1) P=1; order=RL; BF=1	0.0423	+0.0029	+6.9%
short	6	Dynamic (1, 1) P=1; order=LR; BF=1	0.0456	Dynamic (1, 1) P=1; order=RL; BF=1	0.0425	+0.0031	+7.2%
short	8	Dynamic (1, 1) P=1; order=LR; BF=1	0.0451	Dynamic (1, 1) P=1; order=LR; BF=1	0.0438	+0.0014	+3.1%
tiny	2	Dynamic (1, 1) P=1; order=LR; BF=1	0.0026	Dynamic (1, 1) P=1; order=LR; BF=1	0.0017	+0.0009	+50.5%
tiny	4	Dynamic (1, 1) P=1; order=LR; BF=1	0.0026	Dynamic (1, 1) P=1; order=RL; BF=1	0.0017	+0.0009	+51.5%
tiny	6	Dynamic (1, 1) P=1; order=LR; BF=1 * 80% of native samples	0.0026	Dynamic (1, 1) P=1; order=RL; BF=1	0.0017	+0.0009	+51.8%
tiny	8	Dynamic (1, 1) P=1; order=LR; BF=1	0.0026	Dynamic (1, 1) P=1; order=RL; BF=1	0.0017	+0.0009	+52.3%
trip skew	2	Dynamic (1, 1) P=1; order=LR; BF=1 * 80% of native samples	4.3586	Ordinary-first (1, 1) parents=2	4.3446	+0.0140	+0.3%
trip skew	4	Dynamic (3, 3) P=1; order=LR; BF=1	1.8151	Dynamic (3, 3) P=1; order=RL; BF=1	1.9069	-0.0918	-4.8%
trip skew	6	Dynamic (5, 5) P=1; order=LR; BF=1	1.1847	Dynamic (5, 5) P=1; order=LR; BF=1	1.1877	-0.0029	-0.2%
trip skew	8	Dynamic (7, 7) P=1; order=LR; BF=1	1.6814	Dynamic (6, 5) P=1; order=RL; BF=1	1.5908	+0.0906	+5.7%

Policy: (left, right) loop widths; P = parent-worker limit; order = task priority; BF = backfill (1 on, 0 off).

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Model policy vs sweep-best policy — loop workloads (3/3)

Current model after early-diversity change. Gap = 100 × (model time / best time – 1).

Workload	Worker budget	Model allocated policy (modal if varying)	Model time (ms)	Sweep-best policy (remeasured)	Best time (ms)	Extra time (ms)	Gap from best (%)
uneven chunks	2	Dynamic (1, 1) P=1; order=RL; BF=1	0.0693	Dynamic (1, 1) P=1; order=RL; BF=1	0.0683	+0.0010	+1.5%
uneven chunks	4	Dynamic (1, 1) P=1; order=RL; BF=1	0.0691	Dynamic (1, 1) P=1; order=LR; BF=1	0.0679	+0.0012	+1.8%
uneven chunks	6	Dynamic (1, 1) P=1; order=RL; BF=1	0.0692	Dynamic (1, 1) P=1; order=RL; BF=1	0.0680	+0.0012	+1.8%
uneven chunks	8	Dynamic (1, 1) P=1; order=RL; BF=1	0.0699	Dynamic (1, 1) P=1; order=LR; BF=1	0.0691	+0.0009	+1.3%
work skew	2	Dynamic (1, 1) P=2; order=LR; BF=1 * 97% of native samples	6.2934	Dynamic (1, 1) P=1; order=LR; BF=1	6.2945	-0.0011	-0.0%
work skew	4	Dynamic (3, 3) P=1; order=LR; BF=1	2.4131	Dynamic (3, 3) P=1; order=LR; BF=1	2.4060	+0.0071	+0.3%
work skew	6	Dynamic (5, 5) P=1; order=LR; BF=1	1.8211	Dynamic (5, 5) P=1; order=RL; BF=1	1.7011	+0.1200	+7.1%
work skew	8	Dynamic (7, 7) P=1; order=LR; BF=1	1.5439	Dynamic (6, 4) P=1; order=RL; BF=1	1.6835	-0.1396	-8.3%
work skew reversed	2	Dynamic (1, 1) P=2; order=RL; BF=1	6.2044	Ordinary-first (1, 1) parents=2	6.1193	+0.0851	+1.4%
work skew reversed	4	Dynamic (3, 3) P=1; order=RL; BF=1	2.4236	Dynamic (3, 3) P=1; order=RL; BF=1	2.4989	-0.0753	-3.0%
work skew reversed	6	Dynamic (5, 5) P=1; order=RL; BF=1	1.5849	Dynamic (5, 5) P=1; order=RL; BF=1	1.6537	-0.0687	-4.2%
work skew reversed	8	Dynamic (7, 7) P=1; order=RL; BF=1	1.5590	Dynamic (3, 7) P=1; order=RL; BF=1	1.5193	+0.0397	+2.6%

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